

Unsupervised Machine Learning for Muon data quality monitoring in CMS

By: Ava Faubus

Introduction

The Compact Muon Solenoid (CMS) detector plays a pivotal role in refining and testing the Standard Model of Physics, from the discovery of the Higgs boson to the potential to determine the origins of dark matter and better understand the nature of the Higgs. The ongoing impact of CMS is made possible by international efforts to manage and analyze the data coming from CMS. As CMS transitions to the high luminosity era, where significantly more data is expected, advancements in data management are required to sustain the ability of CMS to advance the Standard Model.

In the CMS endcap muon system, the Cathode Strip Chambers (CSCs) require continuous data quality monitoring (DQM) to identify anomalous detector behavior during data collection. AutoDQM was designed as a way to ease the human cost of traditional DQM. Presently, AutoDQM compares runs to a good reference run using chi-squared and max-pull tests, deciding whether to plot the histograms for shifter review based on provided cutoff values for maximum pull and chi squared values. While effective, this method has weaknesses, such as requiring a good representative run to compare to and the prevalence of false anomaly detections. Unsupervised machine learning techniques like principal component analysis (PCA) and autoencoders are an underdeveloped avenue to mitigate these limitations of the current AutoDQM implementation.

Principal component analysis works by analyzing histograms from a subset of runs and determining a set of variables (principal components) that can describe the variation in the data. These principal components can then be used to reconstruct normal or expected distribution to compare runs to, reducing dependence on reference runs. Only good runs can be reconstructed effectively, allowing anomalous runs to be identified by statistical tests performed on the reconstructions. Autoencoders give a nonlinear generalization of this approach using neural network based representations. Both methods are able to handle and adapt to many run conditions, allowing for detection of unprecedented anomalies. Additionally, these anomaly detection techniques are unsupervised, which further cuts down the high human cost of traditional DQM.

Throughout this project, I will train and test existing PCA and autoencoder models for implementation in AutoDQM using offline Run 2 and Run 3 data collected by the CSC subsystem. Throughout this process, I will work out a workflow for the preprocessing of data for this purpose to ensure reproducible results. After training, I will perform a comprehensive comparative analysis between current AutoDQM results and the results from the PCA and

autoencoder methods. This will include characterizing any false positives and ensuring the effectiveness of the updated model against existing DQM metrics. The characterization of false positives will give shifters important insight when reviewing runs marked anomalous by the model, and it paves the way for further updates to AutoDQM.

Software Deliverables

This project will produce software and analysis pipelines for evaluating the performance of anomaly detection methods in CSC data, including:

1. A preprocessing workflow for training and test dataset construction of Run 2 and Run 3 CSC offline data organized into lumisections.
2. Implementing new PCA and autoencoder algorithms into autoDQM
3. Comparative analysis methods for AutoDQM against existing statistical DQM methods, such as max-pull and chi-squared analyses.
4. Documentation and analysis summaries of model performance and false-positive characterization.

Timeline (June 8, 2026 to August 15, 2026)

Weeks 1-2: Dataset construction and preprocessing

I will select good Run 2 and Run 3 offline data and organize it into lumisections. The dataset will be divided into training and testing samples, with fifty percent for model training and fifty percent for evaluating model performance.

Weeks 3-6: Training PCA and autoencoder

Using the constructed dataset, I will train the PCA and autoencoder models for application in anomaly detection in CSC.

Weeks 7-9: Comparative analysis and characterization of false positives

I will compare the performance of the PCA and autoencoder models to the existing statistical methods, particularly max-pull and chi squared tests. My analysis will focus on agreement with the existing statistical methods and characterization of any systematic false positives.

Week 10: Documentation and writeup

I will create a presentation summarizing the project results and detailed documentation describing the developed analysis methods as well as recommendations for future improvements to AutoDQM.